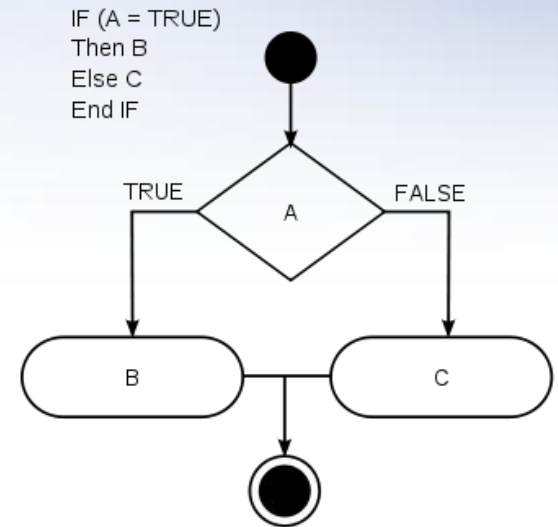


An Overview of Computer Programming

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Topics



- What is a computer?
- What are the components of a computer?
- How does a computer work?
- How does the software work?
- How to write a computer program
- Computer virus
- Q & A



Terminologies



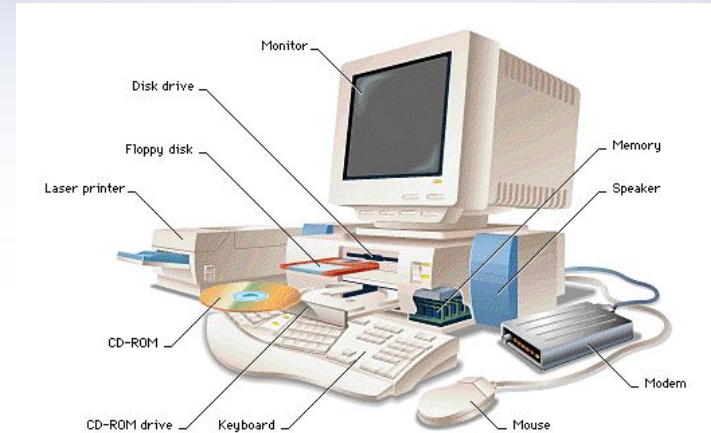
- **Computer** is a programmable machine.
- **Personal computer** (PC) is a computer for an individual user.
- **Computer system** is a collection of *hardware* and *software*.
- **Programming language** is a set of rules, vocabulary and syntax used to instruct the computer to execute certain tasks.
- **Computer program** is a sequence of instructions that can be executed by a computer to perform specific tasks.
- **Programmer** is a person who writes the computer program.
- **Integrated development environment** (IDE) is a set of tools that help you write computer program.
- **Computer virus** is a computer program planted in one computer with the intention of corrupting the recipient computer.

How does a computer work?



•Hardware

- Central processing unit (CPU)
- Memory
- Storage devices
- Input devices
- Output devices



•Software

- Operating System (OS)
- Application programs



How does the software work?



- Software is the computerized instructions that operate the computer, execute particular functions or tasks, and manipulate the data.

For software to perform various functions, it must be programmed.

Choose a programming language



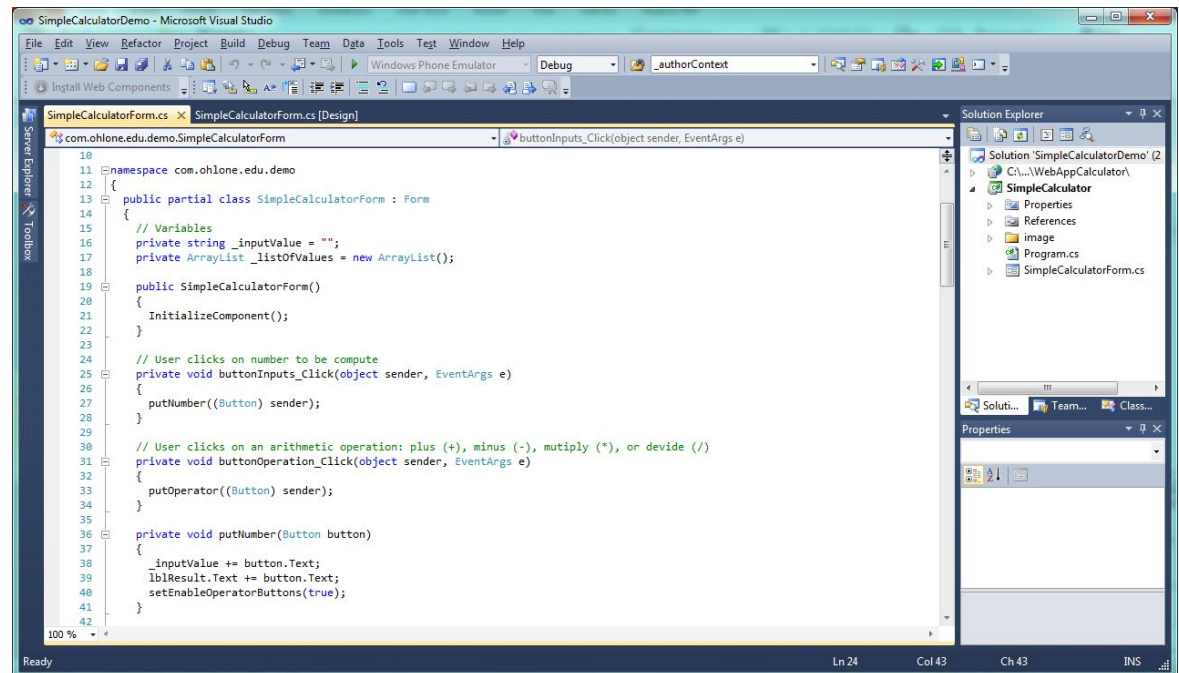
Popular choice of programming languages include:

- C
- C++
- C#
- Java

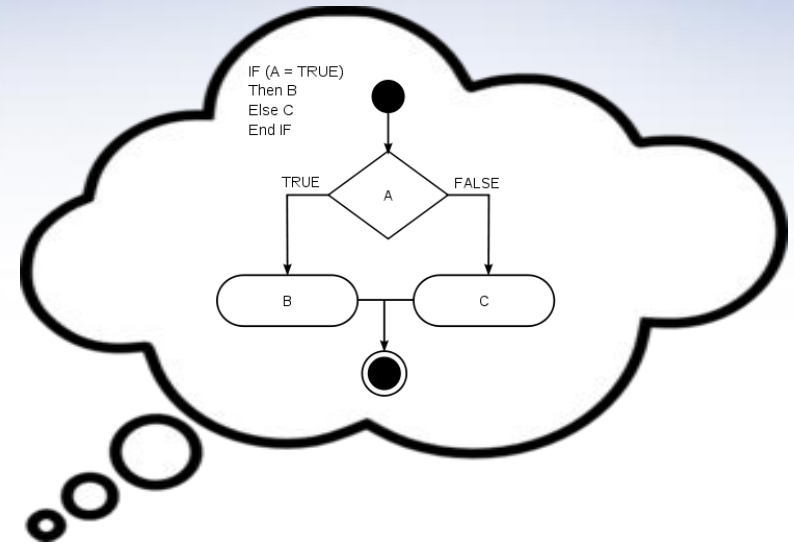
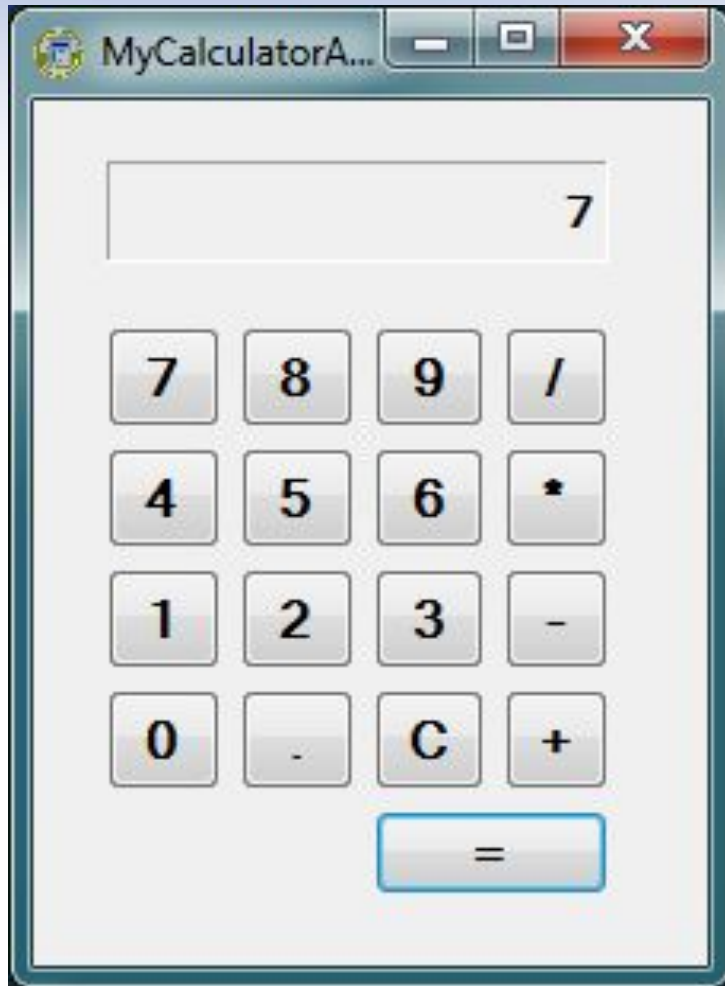
Why should I use an IDE?



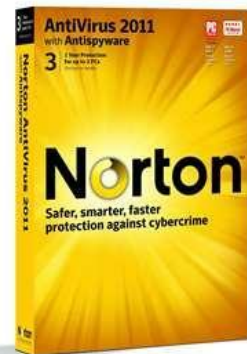
- Integrated Development Environment
 - Code editor, for writing source code
 - Compiler
 - Debugger



How to write a calculator program



What are computer viruses



Summary



- Computer system is made up of hardware and software components.
- Choose a programming language.
- Choose an IDE for writing computer program.
- Protect your personal computer and computer programs from computer viruses.

Q & A



Thank you!